S M Jishanul Islam

CS Master's Student & AI Engineer
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RESEARCH INTERESTS

Multi-modal Learning – Multi-modal Representation – Multi-modal Understanding – Multi-modal AI Applications.

EDUCATION

Dalhousie University, Halifax, Nova Scotia, Canada — 2026 — Present
Master's in Computer Science
Supervisor: Prof. Dr. Frank Rudzicz

United International University, Dhaka, Bangladesh — 2020 — 2024
B.Sc. in Computer Science and Engineering (**Major**: Data Science) — CGPA: 3.77/4.00
Distinction: *Cum Laude* — Major CGPA: 4.00/4.00
Supervisor: Prof. Dr. Swakkhar Shatabda

EXPERIENCE

SAILab, Dalhousie University & Vector Institute — Halifax, NS, Canada
Graduate Student Researcher — September 2026 — Present

- Conducted research focused on hierarchical learning applied to medical domains, supervised by Prof. Dr. Frank Rudzicz.
- Shared research insights in weekly group meetings and collaborated across multiple projects across the team.
- Collaborated across teams from the S AI Lab and the Vector Institute in Toronto.

pipikwan pêhtâkwan — Edmonton, AB, Canada
AI Engineer — October 2025 — Present

- Led the training of the first thinking model catered to serve Indigenous Peoples.
- Built inference pipelines for billion-parameter models to serve large user bases, reducing time-to-first-token to 1s.
- Collaborating with the software engineering team to integrate LLM pipelines to build AI-powered sub-products.

United International University — Dhaka, Bangladesh
Part-time Faculty — October 2025 — July 2026

- Taking core applied CS courses as a faculty from the industry, focusing on algorithms and software engineering.
- Collaborating with the department of CSE to grade, provide feedback, and report student performances and provide industry insights to improve courses.
- Communicating with TAs for course materials creation and course outline building.

Apurba Technologies Ltd. — Dhaka, Bangladesh
Software Engineer — December 2024 — October 2025

- Built scalable and optimized core ML services, improving performance by 10%.
- Co-created the first CLIP-style word recognizer for VLM-based Bengali OCR, recording an accuracy of 95%+.
- Collaborated with stakeholders to implement, research, and complete project deliverables 2-3 weeks before the deadline.

United International University — Dhaka, Bangladesh
Lecturer (contractual) — June 2024 — October 2024

- Taught core CS concepts to students.
- Took exams to monitor students' performance to provide informative feedback, improving average performance by 7%.
- **Courses conducted:** Object-Oriented Programming Lab, Data Structures and Algorithms 1 Lab, and Database Management Systems Theory.

United International University — Dhaka, Bangladesh
Undergraduate Assistant — September 2023 — May 2024

- Managed to facilitate course content and materials with the lab faculty, checked and evaluated assessments, and judged lab projects.
- Created the first course materials for the country's first-ever undergraduate degree in Data Science.
- **Courses supervised:** Programming for Data Science, and Object-Oriented Programming for Data Science.

PUBLICATIONS

Journals

- **Islam, S. M.**, Mustakim, S. H., Hossain, M., Mashira, M., Shourav, N. I., Ahmed, M. R., Islam, S., Islam, A. K. M., & Shatadba, S. "An audio video-based multi-modal fusion approach for speech emotion recognition," **Knowledge-Based Systems (Elsevier)** [Q1; IF 8.0], 2026.

Conferences

- **Islam, S. M.**, Hassan, M., Ovi, M. H., Rabby, A. K. M., & Rahman, F. "BornoDrishti: Leveraging Vision Encoders and Domain-Adaptive Learning for Bangla OCR on Diverse Documents." In the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL) Industry Track. [**ICORE: A**], ACL, 2026.
- Hassan, M., **Islam, S. M.**, Rabby, A. K. M., & Rahman, F. "Beyond Characters: Position-Aware Metric and a Lightweight LLM for Low-Resource Bangla Punctuation Restoration." In 2025 IEEE International Conference on Big Data (BigData). [**ICORE: B**], IEEE, 2025.
- Ahmmed, S., **Islam, S. M.**, Mustakim, S. H., Islam, R., Shanto, S. K., and Islam, S. "IPBlocks: A Blockchain Ecosystem for Secure IP Registration and Decentralized Marketplace." In TENCON 2025 - 2025 IEEE Region 10 Conference. [**ICORE: C**], IEEE, 2025.
- Ahmmed, S., Rahman, T., **Islam, S. M.**, Reyad, A. M., Dey, S., & Farid, D. M. "E-MedViTR: Enhanced Vision Transformers with Registers for Biomedical Image Classification." In 2024 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT). IEEE, 2024.

Workshops

- Mustakim, S. H., **Islam, S. M.**, Muna, U. M., Chowdhury, M., Islam, M. J., Ahmmed, S., Sikder, T., Dhrubo, S. T. A., & Shatabda, S. "Watch, Listen, Understand, Mislead: Tri-modal Adversarial Attacks on Short Videos for Content Appropriateness Evaluation." Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2025, pp. 2955–2964. [**ICORE: A* workshop**]
- **Islam, S. M.**, Mustakim, S. H., Ahmmed, S., Sayeedi, M. F. A., Khandoker, S., Dhrubo, S. T. A., & Hossain, N. "MIMIC: Multimodal Islamophobic Meme Identification and Classification." 3rd Muslims in ML Workshop – NeurIPS, 2024. [**ICORE: A* workshop**]

NOTABLE PROJECTS

NurtureAid (github.com/sadia-ahmmed/AgeWell-Frontend)
Built a real-time AI-based mobile application that simplifies care through caretakers. Led the app's backend and frontend development as a full-stack engineer.
Tech Stack: React Native, Node.js, Flask, MongoDB, Firebase, LLaMA-Index, PyTorch.
Acknowledgements: Champion of the UIU CSE Project Show, Software Engineering Laboratory; Champion of the Hult Prize OnCampus round in UIU; Selected for the Hult Prize Summit in Boston.

Haven (github.com/S-M-J-I/Haven-Harvard-Hsil-Hackathon-2025)
Created a BCI-powered mental health speech-to-speech conversational agent that understands and responds with emotion. Led the app's backend, frontend, and AI development as a full-stack engineer.
Tech Stack: Python, OpenAI, Emotiv, Firebase, Streamlit.
Acknowledgments: Winner, Harvard HSIL Hackathon Dhaka Hub 2024

SKILLS

Programming: C, C++, Java, Python, JavaScript, PHP, SQL, Solidity.
Machine Learning: PyTorch, Tensorflow, Scikit-Learn, Matplotlib, Numpy, Pandas, NLTK, HuggingFace.
Frameworks/Libraries: React.js, Node.js, Spring Boot, FastAPI, Firebase.
DevOps Tools: Git, GitHub, Docker.
Soft Skills: Communication, Leadership, Design Thinking, Public Speaking, Team Player.

ACHIEVEMENTS

Winner, Harvard Health Systems Innovation Lab Hackathon 2025 (Dhaka Hub) 2025
Academic Excellence Scholarship 2020 — 2024
Gold Award (Champion), International Blockchain Olympiad (IBCOL) 2023 2023
Champion, Bhashamul: Bengali Regional Text to IPA National NLP Datathon 2024
Champion, Hult Prize OnCampus Round 2024
Summit Participant, Hult Prize Summit in Boston, USA 2024
Silver Award(1st Runner Up), Bangladesh Blockchain Olympiad (BCOLBD) 2023 2023
1st Runner Up, Intra-University Deep Learning Sprint 2023
Finalist, ICT Innovation Grant (for NurtureAid) 2023
Champion, Software Engineering Lab 2023
Champion, Database Management Systems Lab 2022
Champion, Advanced Object Oriented Programming Lab 2022

EXTRA CURRICULARS

UIU Computer Club - served as the General Secretary 2022 — 2024
Triumph of IBCOL 2023 and the Future of Blockchain in Bangladesh by Jamuna TV - interview link.
Trainer, Hands-on Workshop on Automatic Speech Recognition by Bengali.AI and IUT Computer Society - workshop link.
Trainer, Hands-on Workshop on building a dApp from scratch by UIU Computer Club - workshop link.